



Regulation	:	R2023
Course Name	:	Principles of Artificial Intelligence
Course Code	:	JAL2401
Year / Semester / Branch	:	II/IV/ CSE,CSE (CS),CSE (AI&ML),CS&BS,AI&DS
Unit No	:	I
Unit Title	:	Introduction to Artificial Intelligence and Intelligent Agents
No. of Hours	:	9
Weightage in Exam (%)	:	20%

DEPARTMENT OF COMPUTER SCIENCE AND BUSINESS SYSTEMS

QUESTION BANK

PART A

Q.No.	Questions	Marks	BTL	CO
1	Define Artificial Intelligence. Artificial Intelligence is the simulation of human intelligence in machines that are programmed to think, learn, and problem-solve. It involves tasks such as reasoning, perception, decision-making, and learning from experience.	2	K1	CO1
2	Why is Artificial Intelligence considered an interdisciplinary field? AI is interdisciplinary because it combines knowledge and techniques from various fields such as computer science, mathematics, psychology, neuroscience, linguistics, and	2	K2	CO1

	engineering to develop intelligent systems.			
3	State Turing Test in AI. The Turing Test, proposed by Alan Turing, is a measure of a machine's ability to exhibit intelligent behavior indistinguishable from that of a human. A machine passes the test if a human evaluator cannot reliably distinguish between the machine and a human during a conversation.	2	K1	CO1
4	List out the major branches of AI research. The main branches of AI research include machine learning, natural language processing, robotics, expert systems, computer vision, and cognitive computing.	2	K1	CO1
5	Specify who introduced the concept of artificial intelligence. John McCarthy is often considered the father of artificial intelligence. He coined the term "artificial intelligence" in 1956 and played a major role in the development of AI as a field of study.	2	K1	CO1
6	State the significance of the Dartmouth Conference (1956) for AI. The Dartmouth Conference (1956) is considered the birth of AI as an academic field. It was the first conference where researchers gathered to discuss the possibility of creating machines that could simulate aspects of human intelligence.	2	K1	CO1
7	Outline the recent AI advancement used in real life. Key advancements include the development of deep learning algorithms, natural language processing (such as GPT and BERT), reinforcement learning, AI in healthcare (like diagnostic tools), and autonomous vehicles.	2	K2	CO1
8	Differentiate between Narrow AI and General AI. Narrow AI (or Weak AI) is designed for specific tasks, such as facial recognition or language translation, while general AI (or Strong AI) refers to a hypothetical system with human-like intelligence.	2	K2	CO1
9	How does AI affect job opportunities in different industries? AI can lead to job displacement as automation and intelligent systems replace human workers in various sectors such as manufacturing, retail, and customer service.	2	K2	CO1
10	State the benefits of AI in healthcare. AI can assist in diagnosing diseases more accurately and quickly by analyzing medical data and images, leading to better patient outcomes and more efficient healthcare systems.	2	K1	CO1
11	Define an intelligent agent in AI. An intelligent agent is a system that perceives its environment, processes the information, and takes actions to achieve specific goals, often autonomously. It may involve learning and adaptation.	2	K1	CO1
12	How Siri/Alexa functions as an intelligent agent? A voice assistant like Amazon's Alexa or Apple's Siri is an example of an intelligent agent, as it perceives user inputs (speech), processes them, and takes actions (like setting	2	K2	CO1

	reminders or controlling devices).			
13	Interpret the meaning of “environment” for an intelligent agent. The environment is everything that an intelligent agent interacts with, perceives, and acts upon. It can include physical surroundings, data sources, and other agents.	2	K2	CO1
14	How does an intelligent agent interact with its environment? An intelligent agent interacts with its environment by perceiving information through sensors, processing the information, and then taking actions using actuators or effectors to achieve specific goals.	2	K2	CO1
15	Summarize how an intelligent agent interacts with its environment. Rational behavior refers to an agent making decisions that maximize its expected performance or utility, based on its knowledge of the environment and its goals.	2	K2	CO1
16	How rationality is measured in AI? Rationality is measured by how effectively an agent can achieve its goals, given its knowledge of the environment, available resources, and time constraints.	2	K2	CO1
17	Illustrate the difference between a deterministic and a stochastic environment. A deterministic environment has predictable outcomes for each action taken, while a stochastic environment involves uncertainty, where actions may have unpredictable or probabilistic outcomes.	2	K2	CO1
18	State the two key aspects of a dynamic environment in AI. In a dynamic environment, the environment changes over time (either due to external factors or the agent's own actions), and the agent must continuously adapt to these changes.	2	K2	CO1
19	List out the main components of an intelligent agent. The main components of an intelligent agent are the sensor (to perceive the environment), the actuator (to take actions), and the agent function (to process information and decide what action to take).	2	K1	CO1
20	Outline how does a reflex agent differ from a goal-based agent. A reflex agent selects actions based solely on the current state (using condition-action rules), while a goal-based agent takes into account future consequences and selects actions to achieve specific long-term goals.	2	K2	CO1

PART B

Q.No.	Questions	Marks	BTL	CO
1	Determine the meaning of Artificial Intelligence and explain its key areas and foundations.	16	K3	CO1

2	Summarize the historical development of AI from the 1950s to the present, highlighting key milestones and breakthroughs.	16	K2	CO1
3	Discuss the current advancements in AI, including machine learning, deep learning, NLP, and robotics.	16	K2	CO1
4	Analyze how the use of AI in medical diagnosis influences accuracy, efficiency, and patient care. Break down the potential benefits and examine the risks associated with misdiagnosis, lack of accountability, and over-reliance on technology.	16	K4	CO1
5	Describe an intelligent agent and explain its basic architecture including sensors, actuators, and agent function.	16	K2	CO1
6	Describe the relationship between agents and environments in Artificial Intelligence. What are the types of environments that agents can operate? Provide examples of different environments and discuss how agents adapt to them.	16	K2	CO1
7	Compare and contrast different types of environments in AI and analyze how they influence the rational behavior of an intelligent agent.	16	K4	CO1
8	Apply the concepts of goal-based and utility-based agents to show how they differ, and examine their advantages and limitations in solving AI problems.	16	K3	CO1

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Regulation	:	R2023
Course Name	:	Principles of Artificial Intelligence
Course Code	:	JAL2401
Year / Semester / Branch	:	II/IV/ CSE,CSE (CS),CSE (AI&ML),CS&BS,AI&DS
Unit No	:	II
Unit Title	:	Problem Solving and Searching
No. of Hours	:	9
Weightage in Exam (%)	:	20%

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QUESTION BANK

PART A

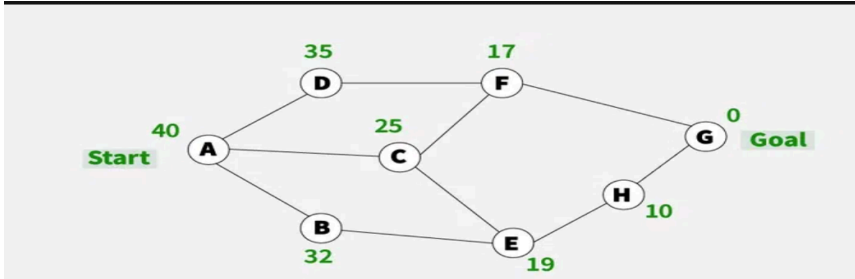
Q.No.	Questions	Marks	BTL	CO
1	Define the term “solving problems by searching” in AI. Solving problems by searching refers to the process of exploring possible solutions to a problem by systematically searching through a state space. AI systems generate a sequence of actions that lead to the desired goal state from an initial state.	2	K1	CO2
2	Discuss the role of problem-solving agent in AI. A problem-solving agent is an intelligent agent that can find a solution to a problem by searching through the problem's state space. It typically involves identifying an initial state, defining a goal state, and using a search algorithm to find a path between these states.	2	K2	CO2
3	Illustrate a problem that can be solved using search techniques. A common example is the 8-puzzle problem , where the agent must find a sequence of moves to move tiles from an initial configuration to a goal configuration.	2	K2	CO2
4	State the role of search algorithms in problem-solving. Search algorithms are used to explore the possible states in a problem space. They help in finding the sequence of actions (or path) that leads from the initial state to the goal state	2	K1	CO2
5	List a few uninformed search strategies used in AI. Uninformed search strategies are algorithms that search the problem space without any domain-specific knowledge, using only the problem definition. Examples include breadth-first search (BFS) and depth-first search (DFS).	2	K1	CO2
6	List down few algorithms that follow uninformed search strategy. 1. Breadth-first search 2. Dijkstra's algorithm or uniform-cost search 3. Depth-first search and the problem of memory 4. Depth-limited and iterative deepening search 5. Bidirectional search	2	K1	CO2
7	Differentiate informed search strategies from uninformed search strategies. Informed search strategies use domain-specific knowledge, typically in the form of heuristics, to guide the search process. A common example is <i>A search*</i> , which uses both the cost of the path and a heuristic estimate of the remaining cost to the goal	2	K2	CO2
8	List down few algorithms that follow uninformed search strategy. 1. Greedy best-first search 2. A* search 3. Search contours	2	K1	CO2

	4. Satisficing search: Inadmissible heuristics and weighted A* 5. Bidirectional heuristic search			
9	Define heuristic function in search algorithms. A heuristic function estimates the cost or distance from a given state to the goal state. It is used to guide the search in informed search strategies like A* search to prioritize the most promising states.	2	K1	CO2
10	How does search change in complex environments? In complex environments, the state space may be large, dynamic, or partially observable, making exhaustive search impractical. Advanced search techniques, such as local search or heuristic search, are often used to handle such complexity.	2	K2	CO2
11	Outline the idea of local search in optimization problems. Local search is a search strategy that explores the problem space by moving from one state to a neighboring state, aiming to find an optimal solution. It is often used for optimization problems like the traveling salesman problem (TSP).	2	K2	CO2
12	How is local search applied in continuous spaces? In continuous spaces, local search algorithms, such as gradient descent, are used to iteratively adjust parameters in small steps to reach a local optimum. These methods are commonly used in machine learning and optimization problems with continuous variables.	2	K2	CO2
13	List the challenges in searching with nondeterministic actions. Nondeterministic actions have unpredictable outcomes, which means the same action in the same state may lead to different results. This introduces uncertainty, and search algorithms must consider all possible outcomes or maintain a belief state to handle this uncertainty.	2	K1	CO2
14	Why search becomes difficult in partially observable environments? In partially observable environments, the agent doesn't have complete information about the current state. This can lead to suboptimal decisions, so search algorithms must use techniques like belief states or probabilistic reasoning to handle uncertainty.	2	K2	CO2
15	How do online search agents handle unknown environments? Online search agents explore an environment without prior knowledge of it. They make decisions based on the current state and the actions they observe, often using strategies like exploration vs. exploitation to balance gathering information and achieving goals.	2	K2	CO2
16	Define state space in AI problem solving. Solving problems by searching refers to the process of exploring possible solutions to a problem by systematically searching through a state space. AI systems generate a sequence of actions that lead to the desired goal state from an initial state.	2	K1	CO2
	Differentiate goal test and path cost in search problems.			

17	A goal test checks whether a given state satisfies the goal condition. Path cost measures the total cost incurred to reach a state from the initial state. While the goal test determines success, path cost helps compare solution quality. Many algorithms aim to minimize the path cost.	2	K2	CO2
18	Why is Breadth-First Search considered complete? Breadth-First Search explores all nodes level by level. It guarantees finding a solution if one exists in a finite state space. Since it examines all shallow nodes before deeper ones, it will not miss a solution. However, it may require large memory.	2	K2	CO2
19	Interpret the importance of heuristics in informed search. Heuristics provide an estimate of how close a state is to the goal. They guide the search toward promising paths, reducing unnecessary exploration. Informed search algorithms like A* use heuristics to improve efficiency. Good heuristics significantly reduce time and space complexity.	2	K2	CO2
20	Discuss the exploration–exploitation trade off in online search? Exploration involves trying new actions to gain information about the environment. Exploitation focuses on using known actions that yield good results. Online search agents must balance both to perform effectively. Too much exploration wastes time, while too much exploitation may miss better solutions.	2	K2	CO2

PART B

Q.No.	Questions	Marks	BTL	CO
1	Discuss the process of solving problems by searching in AI. Explain how the state space, initial state, and goal state are involved in this process.	16	K2	CO2
2	Demonstrate three example problems that can be solved by searching, and apply the concepts of state space, possible actions, and suitable search algorithms for each.	16	K3	CO2
3	Explain A* Algorithm and calculate path cost with suitable example.	16	K5	CO2
4	Analyze how heuristic search strategies improve search performance. Discuss specific informed search algorithms and their applications.	16	K4	CO2
5	Analyze the following graph using Greedy Best First Algorithm.	16	K5	CO2

				
6	Discuss local search techniques in continuous spaces. How do these methods differ from discrete search algorithms, and what challenges arise when dealing with continuous optimization problems?	16	K2	CO2
7	Apply appropriate search algorithms to solve problems involving nondeterministic actions. Demonstrate how these algorithms handle uncertain outcomes with suitable examples and solutions.	16	K3	CO2
8	Evaluate the concept of search with nondeterministic actions. How does it differ from deterministic search?	16	K5	CO2

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Regulation	:	R2023
Course Name	:	Principles of Artificial Intelligence
Course Code	:	JAL2401
Year / Semester / Branch	:	II/IV/ CSE,CSE (CS),CSE (AI&ML),CS&BS,AI&DS
Unit No	:	III

Unit Title	:	Constraint Satisfaction Problems and Game Theory
No. of Hours	:	9
Weightage in Exam (%)	:	20%

DEPARTMENT OF COMPUTER SCIENCE AND BUSINESS SYSTEMS

QUESTION BANK

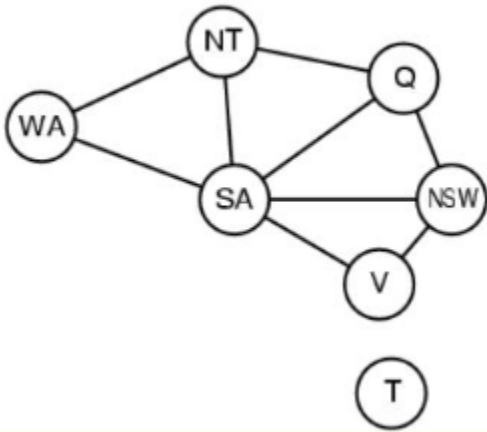
PART A

Q.No.	Questions	Marks	BTL	CO
1	Define Constraint Satisfaction Problem. A Constraint Satisfaction Problem (CSP) is a problem where the goal is to find values for a set of variables, subject to a set of constraints that restrict the possible values these variables can take. CSPs are widely used in areas like scheduling, planning, and resource allocation. <i>Examples:</i> Sudoku, map coloring problem, N-queens problem.	2	K1	CO3
2	Mention few examples for Constraint Satisfaction Problem (CSP). • Sudoku • Map coloring problem • Cryptarithmic puzzles • N-queens problem • Crossword puzzle • Scheduling problems • Job-Shop Scheduling • Timetable Scheduling	2	K1	CO3
3	Outline the components of a Constraint Satisfaction Problem. The components of a CSP include: • Variables: A set of variables to be assigned values. • Domains: A set of possible values for each variable. • Constraints: Rules that specify allowable combinations of variable assignments.	2	K2	CO3
4	List down the techniques applied to solve CSPs. • Backtracking	2	K1	CO3

	• Forward Checking • Constraint Propagation (Arc Consistency, AC-3 Algorithm) • Local Search (Min-Conflicts Algorithm)			
5	Discuss constraint propagation in the context of CSPs. Constraint propagation is the process of reducing the search space by using the constraints to infer additional information about variable assignments. This can simplify the problem and make solving more efficient	2	K2	CO3
6	Discuss backtracking search in solving CSPs. Backtracking search is a systematic search algorithm for solving CSPs. It assigns values to variables one by one and backtracks when a constraint is violated, returning to previous variables to try alternative assignments.	2	K2	CO3
7	How is local search applied to Constraint Satisfaction Problems? Local search techniques for CSPs involve starting with an initial solution and iteratively improving it by making small changes, such as assigning new values to variables. Common methods include hill climbing and simulated annealing.	2	K2	CO3
8	Summarize the importance of problem structure in solving CSPs. The structure of a problem affects the choice of solving methods. Problems with high constraint density or certain symmetries may benefit from techniques like constraint propagation, while others might be better suited to backtracking or local search.	2	K2	CO3
9	State adversarial search in the context of game theory. Adversarial search refers to the process of finding optimal strategies for two or more agents (players) with conflicting goals. It is commonly used in game theory to model competitive situations, such as chess or tic-tac-toe.	2	K1	CO3
10	Define game theory in the context of AI. Game theory is a mathematical framework used to study decision-making in situations where multiple agents make choices that affect each other's outcomes. It helps determine optimal strategies in competitive and cooperative environments.	2	K1	CO3
11	Outline optimal decisions in the context of game playing? Optimal decisions in game playing refer to moves that maximize a player's chance of winning, assuming both players play optimally. This typically involves calculating the best possible outcome using search algorithms like Minimax.	2	K2	CO3
12	Discuss heuristic Alpha-Beta pruning in game search. Alpha-Beta pruning is an optimization technique for the Minimax algorithm. It eliminates branches of the search tree that do not need to be explored because they cannot affect the final decision, improving efficiency without affecting the outcome.	2	K2	CO3
13	Define Monte Carlo Tree Search (MCTS). Monte Carlo Tree Search is a search algorithm used for decision-making in games. It uses random simulations to estimate the value of different moves and selects the most promising one	2	K1	CO3

	based on statistical outcomes.			
14	Interpret the significance of stochastic game in AI. A stochastic game is a type of game in which outcomes are partly random. Players' actions lead to probabilistic state transitions, and strategies must take into account both the randomness and the adversarial nature of the game.	2	K2	CO3
15	Distinguish partially observable games from fully observable games. In partially observable games, players do not have complete information about the game state. They must rely on incomplete observations and make decisions based on available information, unlike fully observable games where all players know the state.	2	K2	CO3
16	Mention few limitations of game search algorithms. Few limitations include: • Computation complexity: As game trees grow exponentially, exhaustive search becomes impractical. • Imperfect information: In games with incomplete or imperfect information, finding optimal strategies is challenging. • Adversarial nature: In two-player games, the opposing player's strategy needs to be considered, which increases the difficulty of finding optimal decisions.	2	K1	CO3
17	Define partially observable game. Give an example. A partially observable game is one where players do not have full knowledge of the game state. Players can only observe parts of the game state, leading to uncertainty in decision-making. Poker is an example of a partially observable game, where players cannot observe the cards of their opponents, only their own cards and the community cards.	2	K1	CO3
18	Discuss the concept of "information set" in partially observable games. An information set is a collection of game states that are indistinguishable to a player due to lack of full observation. Players make decisions based on their information set rather than the full game state.	2	K2	CO3
19	How does the lack of full observation affect decision-making in partially observable games? The lack of full observation forces players to make decisions based on incomplete information, relying on strategies like probabilistic reasoning and inference to guess the hidden elements of the game state.	2	K2	CO3
20	How does Monte Carlo Tree Search (MCTS) handle uncertainty in partially observable games? MCTS is a search algorithm that uses random simulations to explore possible future states. In partially observable games, it handles uncertainty by simulating possible future states based on belief states, reducing reliance on complete state information.	2	K2	CO3

PART B

Q.No.	Questions	Marks	BTL	CO
1	Apply the concept of Constraint Satisfaction Problem(CSP) to solve a map-coloring problem using the following constraint (i) The goal is to assign colors to each region so that no neighboring regions have the same color.  (ii) The map-coloring problem represented as a constraint graph 	16	K3	CO3
2	Analyze how a Constraint Satisfaction Problem (CSP) is defined using variables, domains, and constraints. Analyze the factors to consider when modeling a problem as a CSP. Also, analyze the N-Queens problem by showing its CSP representation and how constraints reduce the solution space.	16	K4	CO3
3	Implement the backtracking search algorithm to solve a Constraint Satisfaction Problem (CSP). Explain the approach to solve the 8-Queens problem and illustrate the steps to obtain a valid solution.	16	K3	CO3
4	Evaluate the effectiveness of local search techniques in solving Constraint Satisfaction Problems (CSPs). Assess the principles of local search algorithms and justify their suitability for solving problems such as the map-coloring problem.	16	K5	CO3
5	Apply adversarial search techniques to model competitive scenarios in AI, such as games. Demonstrate how algorithms like Minimax and	16	K3	CO3

	Alpha–Beta pruning can be used to determine optimal strategies in such environments.			
6	Apply the concepts of game theory to model and solve problems in competitive environments in AI. Use suitable examples to demonstrate how game theory can be used for decision-making in such scenarios.	16	K3	CO3
7	Evaluate the effectiveness of heuristic Alpha–Beta pruning in game trees for optimizing the Minimax algorithm. Assess how it improves decision-making efficiency in game-playing AI, and justify its use by analyzing its advantages and limitations with a suitable example.	16	K5	CO3
8	Analyze the Monte Carlo Tree Search (MCTS) algorithm and its role in decision-making for game-playing AI. Analyze how the algorithm operates by breaking down its key steps.	16	K4	CO3

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Course Name	:	Principles of Artificial Intelligence
Course Code	:	JAL2401
Year / Semester / Branch	:	II/IV/ CSE,CSE (CS),CSE (AI&ML),CS&BS,AI&DS
Unit No	:	IV
Unit Title	:	Knowledge and Reasoning
No. of Hours	:	9
Weightage in Exam (%)	:	20%

DEPARTMENT OF COMPUTER SCIENCE AND BUSINESS SYSTEMS

QUESTION BANK

PART A

Q.No.	Questions	Marks	BTL	CO
1	Define fuzzy logic. Fuzzy logic is a form of logic that deals with reasoning that is approximate rather than fixed and exact. It allows for degrees of truth between 0 and 1. Fuzzy sets allow partial membership. Eg. If "tall" is a fuzzy set, a height of 5'9" might have a membership of 0.6, while 6'3" might have 0.9.	2	K1	CO4
2	Discuss the First-Order Logic (FOL). First-Order Logic (FOL) is a formal system used for representing knowledge in AI, where statements can be made about objects and	2	K2	CO4

	their relationships using quantifiers, variables, predicates, and constants.			
3	List down the main components of First-Order Logic. The main components are: Objects: Individuals in the domain. Predicates: Relations between objects. Quantifiers: Universal ($\forall$) and existential ($\exists$). Functions: Map objects to objects.	2	K1	CO4
4	Discuss the syntax of First-Order Logic. The syntax of FOL defines how valid sentences can be constructed. It includes: Constants, variables, predicates, functions, and logical connectives. Quantifiers: Universal ($\forall$) and existential ($\exists$).	2	K2	CO4
5	State the semantics of First-Order Logic. The semantics of FOL refers to the meaning of its expressions. It defines how sentences in FOL are interpreted in terms of objects, functions, relations, and domains.	2	K1	CO4
6	Discuss the role of knowledge representation in the context of AI. Knowledge representation is the way in which knowledge about the world is structured and stored in a form that an AI system can use to reason and make decisions.	2	K2	CO4
7	Define quantifiers. What are the different quantifiers used in FOL? Quantifiers are symbols used to express properties of entire collections of objects, instead of enumerating the objects by name. FOL contains two standard quantifiers: - Universal quantifiers $\forall$ - "for all" – Eg. $\forall x \text{ King}(x)$ - "For all x, if x is a king, then x is a person." - Existential quantifiers $\exists$ - "for some x" –Eg. $\exists x \text{ Crown}(x) \wedge \text{OnHead}(x, \text{John})$ - "John has a crown on his head".	2	K1	CO4
8	Illustrate the predicate logic for the following statements: i) Every parent loves their child. ii) There exists a person who has no children $\forall x, y \text{ Parent}(x) \wedge \text{Child}(y, x) \rightarrow \text{Loves}(x, y)$ $\exists x, y \text{ Parent}(x) \rightarrow \neg \text{Child}(x, y)$	2	K2	CO4
9	Summarize Knowledge Engineering process in First-Order Logic. Knowledge engineering involves the creation, structuring, and encoding of knowledge into a form that AI systems can use, typically through the use of formal languages like First-Order Logic.	2	K2	CO4
10	List out the steps in knowledge engineering process. The process of construction of knowledge-base is called as knowledge engineering. The process has the following steps: - Identify the questions - Assemble the relevant knowledge - Decide on a vocabulary of predicates, functions, and constants	2	K1	CO4

	4. Encode general knowledge about the domain 5. Encode a description of the problem instance 6. Pose queries to the inference procedure and get answers 7. Debug and evaluate the knowledge base					
11	Explain the difference between Propositional Logic and First-Order Logic. Propositional logic represents statements as variables that can either be true or false, while FOL includes variables that can refer to objects and predicates that express relations between them, providing a richer language.	2	K2	CO4		
12	List out the Limitations of First-Order Logic. 1. Computational Complexity 2. Expressiveness vs. Decidability 3. Handling Uncertainty	2	K2	CO4		
13	How does unification assist in First-Order inference? Unification is used in First-Order inference to match terms in a logical expression, allowing for the application of inference rules to derive new facts.	2	K2	CO4		
14	State forward and backward chaining in First-Order Logic. Forward chaining is a data-driven inference technique where reasoning starts from known facts and applies rules to derive new facts until the goal is achieved. Backward chaining is a goal-driven inference technique where reasoning starts from a goal and works backward to find the facts that support the goal.	2	K1	CO4		
15	Outline uncertainty and causes for uncertainty. Uncertainty is a situation when an agent may never know for sure what state it is in now or where it will end up after a sequence of actions. It is caused due to partial observability, nondeterminism, or adversaries.	2	K2	CO4		
16	State the Bayes' Rule in probability theory. Bayes' Rule describes the relationship between conditional probabilities, allowing for the update of probabilities based on new evidence. It is written as $P(Y X) = \frac{P(X Y)P(Y)}{P(X)}$	2	K1	CO4		
17	Differentiate forward chaining and backward chaining in First-Order Logic.			2	K2	CO4
	Feature	Forward Chaining	Backward Chaining			
	Definition	Starts with known facts and applies inference rules to derive new facts until the goal is reached.	Starts with the goal (query) and works backward to find supporting facts.			
	Purpose	Useful when deriving multiple conclusions from known data.	More efficient when verifying a specific goal.			
	Direction of Reasoning	Data-driven (bottom-up approach).	Goal-driven (top-down approach).			

	Efficiency	Can be inefficient if many irrelevant facts are generated before reaching the goal.	More efficient when searching for specific answers but can fail if the search space is large.			
18	Define Naive Bayes model. The Naive Bayes model is a probabilistic classifier that assumes independence between the features in the data, and it uses Bayes' Rule to calculate the probability of a class given the feature values.			2	K1	CO4
19	Discuss the advantages of using Bayes' Rule in AI. Bayes' Rule allows for dynamic updating of beliefs based on new evidence, making it useful in decision-making and probabilistic reasoning in uncertain environments.			2	K2	CO4
20	How can a naive Bayes model be used for the task of text classification? Text classification is the process of assigning a given text to one or more predefined set of categories based on its content.			2	K2	CO4

PART B

Q.No.	Questions	Marks	BTL	CO
1	Demonstrate the syntax and semantics of First-Order Logic. How do they work together to represent knowledge?	16	K3	CO4
2	Investigate the role of each step in the Knowledge Engineering process in AI and its relationship with First-Order Logic.	16	K4	CO4
3	Analyze the challenges of using First-Order Logic for knowledge representation in complex environments and compare it with Propositional Logic in solving AI problems.	16	K4	CO4
4	Recommend where Propositional Logic and First-Order Logic are best applied for problem-solving, based on their expressiveness.	16	K5	CO4
5	Examine the process of unification in First-Order Logic and its significance in inference.	16	K4	CO4
6	Apply Bayes' rule to explain the Naive Bayes model and state where it is most effective.	16	K3	CO4
7	An AI knowledge base contains the following statements: $\forall x (\text{Scientist}(x) \rightarrow \exists y \text{Researches}(x,y))$ $\exists x (\text{Scientist}(x) \wedge \neg \text{Researches}(x, \text{AI}))$ $\forall x (\text{Researches}(x, \text{AI}) \rightarrow \text{Innovator}(x))$ Tasks: (i) Evaluate whether the knowledge base is logically consistent. (ii) Determine if all scientists can be considered innovators. (iii) Critically assess how universal and existential quantifiers influence the reasoning outcome.	16	K5	CO4
8	Analyze the mechanisms of forward and backward chaining in AI, including their implementation, differences, and suitability for various problem-solving scenarios.	16	K4	CO4

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Regulation	:	R2023
Course Name	:	Principles of Artificial Intelligence
Course Code	:	JAL2401
Year / Semester / Branch	:	II/IV/ CSE,CSE (CS),CSE (AI&ML),CS&BS,AI&DS
Unit No	:	V
Unit Title	:	Expert Systems and Applications Of Artificial Intelligence
No. of Hours	:	9
Weightage in Exam (%)	:	20%

DEPARTMENT OF COMPUTER SCIENCE AND BUSINESS SYSTEMS

QUESTION BANK

PART A

Q.No.	Questions	Marks	BTL	CO										
1	Define expert system. Give some examples. Expert systems are computer programs that are derived from a branch of computer science research called AI. The programs that achieve expert level competence in solving problems in task areas by bringing to bear a body of knowledge about specific tasks are called expert systems or knowledge base. Examples: • DENDRAL – used in chemical mass spectroscopy to identify chemical constituents • MYCIN – medical diagnosis of illness • DIPMETER – geological data analysis for oil • PROSPECTOR – geological data analysis for minerals • XCON/R1 – configuring computer systems	2	K1	CO5										
2	Differentiate between Knowledge Base and Inference Engine. <table> <tr> <th>Knowledge Base</th> <th>Inference Engine</th> </tr> <tr> <td>It is the repository of facts, rules, and domain-specific information.</td> <td>It is the reasoning component of the expert system.</td> </tr> <tr> <td>It is analogous to the expert's knowledge.</td> <td>It processes the information in knowledge base to solve problems and make decisions.</td> </tr> <tr> <td>It stores what experts know about the problem domain.</td> <td>It emulates the cognitive processes that human experts use to arrive at conclusions.</td> </tr> <tr> <td>It is a set of sentences that represents some assertions about the world.</td> <td>It engine enables the expert system to draw deductions from the rule in knowledge base.</td> </tr> </table>	Knowledge Base	Inference Engine	It is the repository of facts, rules, and domain-specific information.	It is the reasoning component of the expert system.	It is analogous to the expert's knowledge.	It processes the information in knowledge base to solve problems and make decisions.	It stores what experts know about the problem domain.	It emulates the cognitive processes that human experts use to arrive at conclusions.	It is a set of sentences that represents some assertions about the world.	It engine enables the expert system to draw deductions from the rule in knowledge base.	2	K2	CO5
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3	List down the characteristics of expert systems. • Operates as an interactive system • Tools have the ability to filter knowledge • Make logical inferences based on knowledge stored • Ability to explain reasoning • Domain-specific • Capability to assign confidence values • Cost-effective alternative to human expert	2	K1	CO5										
4	List out the roles of individuals who interact with the expert system. Domain expert: The individuals who currently are experts in solving the problems. Knowledge engineer : The individual who encodes the expert's knowledge in a declarative form that can be used by the expert system.	2	K2	CO5										

	User : The individual who will be consulting with the system to get advice which would have been provided by the expert.			
5	Outline the advantages of Expert system. Availability: Expert systems are available easily due to mass production software. • Cheaper: The cost of providing expertise is not expensive. • Reduced danger: They can be used in any risky environments where humans cannot work with. • Permanence: The knowledge will last long indefinitely. • Multiple expertise: It can be designed to have knowledge of many experts.	2	K2	CO5
6	Discuss the roles of Expert Systems. • Advising • Instructing and assisting human in decision making • Demonstrating • Deriving a solution • Diagnosing • Explaining • Interpreting input • Predicting results • Justifying the conclusion • Suggesting alternative options to a problem	2	K2	CO5
7	List out the limitations of Expert System. • Not widely used or tested • Limited to relatively narrow problems • Cannot readily deal with “mixed” knowledge • Possibility of error • Cannot refine own knowledge base • Difficult to maintain • May have high development costs • Raise legal and ethical concerns	2	K1	CO5
8	Summarize the applications of Expert Systems. • Credit granting • Disease diagnosis and medicine prescription • Information management and retrieval • AI and expert systems embedded in products • Plant layout design • Hospitals and medical facilities • Help desks and assistance • Employee performance evaluation • Loan analysis	2	K2	CO5
9	Define knowledge acquisition. Knowledge acquisition refers to the process of gathering,	2	K1	CO5

	synthesizing, and translating expert knowledge from domain experts into a format that can be used by intelligent systems, such as expert systems or machine learning algorithms.			
10	How does MYCIN help in identifying the bacterial infection? MYCIN was an early expert system that used artificial intelligence to identify bacteria causing severe infections, such as bacteria and meningitis, and to recommend antibiotics, with the dosage adjusted for patient's body weight — the name derived from the antibiotics themselves, as many antibiotics have the suffix "-mycin". The MYCIN system was also used for the diagnosis of blood clotting diseases.	2	K2	CO5
11	Define DART. The Dynamic Analysis and Replanning Tool, commonly abbreviated to DART, is an artificial intelligence program used by the U.S. military to optimize and schedule the transportation of supplies or personnel and solve other logistical problems. DART uses intelligent agents to aid decision support systems located at the U.S. Transportation and European Commands	2	K1	CO5
12	Define XCON. XCON (eXpert CONfigurer) is a production-rule-based system designed to automatically select the necessary components for VAX computer systems based on customer orders because the potential configurations of VAX components was too numerous for humans to calculate efficiently, leading to errors in orders.	2	K1	CO5
13	Summarize the application domains of Artificial Intelligence. - Healthcare - Disease diagnosis, drug discovery, personalized treatment, virtual health assistants - Cybersecurity - Threat detection, AI-driven fraud prevention, secure authentication. - Education - AI tutors, personalized learning, automated grading, smart content generation. - Agriculture - Precision farming, crop health monitoring, automated harvesting.	2	K2	CO5
14	How is AI used in environmental management? AI-driven tools facilitate a wide range of environmental monitoring activities, including the following - AI for Cleaner Air – Smart Pollution Control - AI for Water Preservation – Smart Water Management - AI for Better Agriculture – Smart Farming - AI for Better e-Waste Management – Smart Monitoring/Control - AI for Climate Control – Smart Energy Optimization	2	K2	CO5
15	Interpret the role of AI in systems biology. By simulating the behavior of a biological system under various conditions, AI models can predict how the system will respond to	2	K2	CO5

	environmental changes, such as the introduction of a new drug. In summary, AI can develop models that simulate the behavior of biological systems at a molecular level.			
16	How AI is applied in driving agricultural innovation? AI provides local farmers with access to a wealth of information and tools, enabling them to optimize their yields and resource utilization and to make more informed decisions about their farming practices including insights into optimal planting times, soil health management, efficient water usage, and pest control strategies.	2	K2	CO5
17	How is AI used in medical diagnosis? AI systems have been used to analyze medical images, such as those collected via X-rays or MRI scans. These tools help in diagnosing conditions like cancer, retinal diseases, pneumonia, and more. Example: AI diagnostic assistance in cardiology, where deep learning algorithms are used to diagnose heart attacks, like how cardiologists would. In other cases, AI networks have been trained with clinical images to assist with dermatological diagnoses, accurately classifying skin lesions.	2	K2	CO5
18	Explain the incredible ways the smartphone can help track health and disease. a) A smartphone test for female fertility b) Type 2 diabetes c) Pancreatic Cancer d) Skin Cancer e) HIV and syphilis	2	K2	CO5
19	Mention few machine learning algorithms used in in the diagnosis of various diseases.	2	K2	CO5
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20	Summarize the future challenges in Artificial Intelligence. - Understanding and mimicking complex human cognitive functions. - Building robots that can truly simulate human-like behaviors in real-world situations. - Programming machines to feel or respond emotionally like humans. - The impact of AI on ethical dilemmas, such as job displacement, decision accountability, and privacy concerns.	2	K2	CO5

PART B

Q.No.	Questions	Marks	BTL	CO
1	Apply the architecture of an expert system to design a decision-support system for medical diagnosis. Explain in detail the components such as knowledge base, inference engine, and user interface, and illustrate their interaction with a neat diagram.	16	K3	CO5
2	Demonstrate the working of expert systems such as MYCIN, DART, or XCON. Explain their problem domain, knowledge representation, and inference mechanism with suitable examples.	16	K3	CO5
3	Analyze the roles, characteristics, and limitations of expert systems. Discuss how these factors affect their performance and reliability in real-world applications.	16	K4	CO5
4	Evaluate the effectiveness of expert systems compared to human experts in terms of accuracy, cost, consistency, and scalability. Justify your answer with suitable examples.	16	K5	CO5
5	Apply Artificial Intelligence techniques to solve environmental management problems such as pollution control, climate monitoring, and smart agriculture.	16	K3	CO5
6	Determine the application of AI in healthcare and disease diagnosis, including medical imaging, predictive analytics, and smartphone-based diagnostic systems.	16	K3	CO5
7	Examine the impact of Artificial Intelligence in systems biology, agriculture, and biomedical applications, highlighting benefits and limitations.	16	K4	CO5
8	Assess the future challenges and ethical issues of AI applications in industries such as healthcare, transportation, and environmental systems, providing a critical evaluation.	16	K5	CO5

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